INNOVATELA 2026 / PORTS & LOGISTICS

HarborGuard

See underwater risk. Keep people in control.

An explainable digital twin for port security and disaster recovery.

Everyday security

Investigate underwater activity and compare targeted response options.

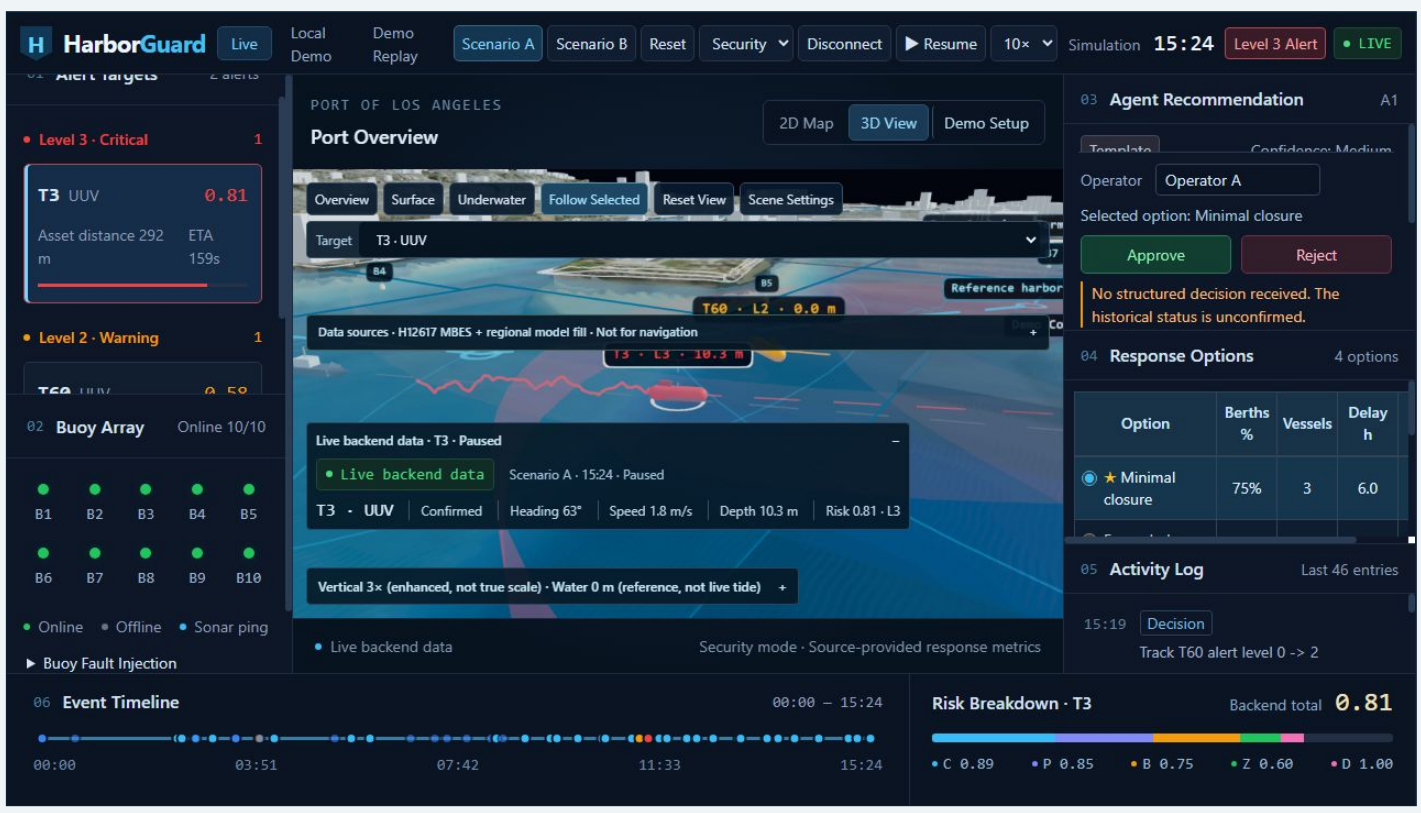
Disaster recovery

Prioritize surveys and candidate routes without erasing unresolved observations.

FROM OBSERVATION TO HUMAN DECISION

01 Observe	02 Track + score	03 Compare	04 Explain	05 Approve
Synthetic acoustics	Tracks and 3 alert levels	Response options	Template Agent	Operator acts first

3D DIGITAL TWIN / REAL BACKEND SIMULATION



Actual A/B/C browser capture, paused at 15:24 simulation time.
Synthetic observations and AIS; archived terrain, not navigation clearance.

ACTUAL SCENARIO A

L3 CRITICAL

L2 WARNING

5 tracks / 10 buoys

T3 risk: 0.81

YOLO26n on synthetic sonar.
Template Agent: opt-1.
Residual risk: 0.55.
Operator review required.

NO AUTO APPROVAL

Alerts are model outputs, not truth.

TWIST 01 / PROPOSED FUNDING

Federal R&D pathway

NSF SBIR Phase I

Real-echo validation and calibrated uncertainty across noisy sites.
Project Pitch, then invited proposal.

Eligibility review required. No award secured.

TWIST 02 / GLOBAL ADAPTATION

Local data. Same human control.

Los Angeles -> Shanghai

Local maps, depths and tide datum.
Recalibrate sensors and detection.
Mandarin UI and local data controls.
Local authorities retain approval.

Proposed Yangshan pilot; no overseas deployment.

2 alerts	4 options	46 logs
Critical + warning	Compare response impact	Agent + activity trail
Observed in this isolated run	Human approval remains required	Actual backend records at 15:24

RESPONSIBLE USE

Could be misused for surveillance or unsafe operational decisions. Human approval and explicit uncertainty limit risk; this prototype is not certified for navigation or port reopening.

Capture: d2d4c6f + English display text | 17 Oct 2026 | Isolated local simulation.
Unresolved recovery observations remain unsurveyed. Real marine validation remains open.

Funding: NSF 26-510 Applicant eligibility

github.com/ChrisJiang123/harborguard

CODE + RUNBOOK